



Lakshya JEE 2.0 (2024)

Differential Equations

DPP-04

1. Solve the differential equation $x \frac{dy}{dx} + y = x^3 y^6$.

(1) $y^{-5} \cdot x^{-5} = \frac{5}{2} x^{-2} + C$

(2) $y^{-4} \cdot x^{-4} = \frac{5}{2} x^{-2} + C$

(3) $y^{-3} \cdot x^{-3} = \frac{5}{2} x^{-2} + C$

(4) None of these

2. Find the solution of differential equation

$$\frac{dy}{dx} - y \tan x = -y^2 \sec x.$$

(1) $y^3 |\sec x| = \tan x + c$

(2) $y^2 |\sec x| = \tan x + c$

(3) $y^{-1} |\sec x| = \tan x + c$

(4) $y |\sec x| = \tan x + c$

3. The solution of the differential equation

$$y dx + (x + x^2 y) dy = 0$$

(1) $\log y = Cx$

(2) $-\frac{1}{xy} + \log y = C$

(3) $\frac{1}{xy} + \log y = C$

(4) $-\frac{1}{xy} = C$

4. $\frac{dy}{dx} = \frac{e^y}{x^2} - \frac{1}{x}$, then:

(1) $2xe^{-y} = 1 - kx^2$

(2) $2xe^{-x} = 1 + kx^2$

(3) $2xe^{-y} = 1 + kx$

(4) $2xe^{-y} = 1 + kx^2$

5. $\frac{dy}{dx} = \frac{y^2 - x}{2y(x+1)}$, then:

(1) $y^2 = (x+1) \log \left(\frac{C}{x+1} \right) - 1$

(2) $y^2 = (x+1)^2 \log \left(\frac{C}{x+1} \right) - 1$

(3) $y^2 = (x+1)^3 \log \left(\frac{C}{x+1} \right) - 1$

(4) $y^2 = (x+1)^{-1} \log \left(\frac{C}{x+1} \right) - 1$

6. Solve: $\frac{dy}{dx} (x^2 y^3 + xy) = 1$

(1) $xy^2 = e^{\frac{x^2}{2}} + C$

(2) $e^{\frac{x^2}{2}} = x \cdot e^{\frac{y^2}{2}} + C$

(3) $e^{\frac{y^2}{2}} = (y^2 - 1) + C$

(4) $-\frac{e^{\frac{y^2}{2}}}{x} = (y^2 - 1) e^{\frac{y^2}{2}} + C$

7. Solve: $\frac{dy}{dx} + \frac{y}{x} \cdot \log y = \frac{y}{x^2} (\log y)^2$

(1) $\log y = \frac{x}{y^2} + C$

(2) $\frac{1}{x \log y} = \frac{1}{2x^2} + C$

(3) $x \log y = 2x^2 + C$

(4) None of these

8. Solve: $\frac{dy}{dx} + x(x+y) = x^3(x+y)^3 - 1$.

(1) $\frac{y}{x} = e^{x^2} + C$

(2) $\frac{y}{x} = (x^2 + 1)^2 + C.e^{x^2}$

(3) $\frac{1}{(x+y)^2} = (x^2 + 1) + C.e^{x^2}$

(4) $\frac{1}{(x+y)^2} = (x+1) + C.e^x$

9. The solution of the differential equation $\frac{dy}{dx} = \frac{y \cos x - y^2}{\sin x}$ is equal to (where c is an arbitrary constant)

(1) $\sin x = x - y + c$

(2) $\sin x = x + y + c$

(3) $\sin x = xy + cy$

(4) $\frac{\sin x}{x} = y + c$

10. If $\frac{dy}{dx} = \frac{y^3}{e^{2x} + y^2}$ and $y(0) = 1$, then:

(1) $y^2 = e^{2x} - 2e^{2x} \ln y$

(2) $y^2 = e^{2x} + 2e^{2x} \ln y$

(3) $y^2 = e^{2x} - \frac{1}{2}e^{2x} \ln y$

(4) $y^2 = e^{2x} - \frac{1}{2}e^{2x} \ln y$

11. Let $f : [1, \infty) \rightarrow [2, \infty)$ be a differentiable function such that $f(1) = 2$. If $6 \int_1^x f(t) dt = 3x f(x) - x^3$ for all $x \geq 1$, then the value of $f(2)$ is

12. The solution of the differential equation $\frac{dy}{dx} = \frac{x-y+3}{2(x-y)+5}$ is

(1) $2(x-y) + \log(x-y) = x + c$

(2) $2(x-y) - \log(x-y+2) = x + c$

(3) $2(x-y) + \log(x-y+2) = x + c$

(4) None of these

13. The general solution of the differential equation $(2x - y + 1)dx + (2y - x + 1)dy = 0$ is

(1) $x^2 + y^2 + xy - x + y = c$

(2) $x^2 + y^2 - xy + x + y = c$

(3) $x^2 - y^2 + 2xy - x + y = c$

(4) $x^2 - y^2 - 2xy + x - y = c$

14. The equation of the curve passing through the point $(1, 1)$ and satisfying the differential equation $\frac{dy}{dx} = \frac{x+2y-3}{y-2x+1}$ is

(1) $x^2 - 4xy - y^2 + 6x + 2y - 4 = 0$

(2) $x^2 + 4xy - y^2 - 6x + 2y + 4 = 0$

(3) $x^2 + 4xy - y^2 - 6x - 2y + 4 = 0$

(4) $x^2 + 4xy + y^2 - 6x - 2y - 4 = 0$

15. Solve the DE $\frac{dy}{dx} = \frac{x+2y-1}{x+2y+1}$.

(1) $y + x - \frac{2}{3} \ln(3x + 6y - 1) = c$

(2) $x + y + \frac{2}{3} \ln(3x + 6y - 1) = c$

(3) $x - y + \frac{2}{3} \ln(3x + 6y - 1) = c$

(4) $y - x + \frac{2}{3} \ln(3x + 6y - 1) = c$

16. Solve: $\frac{dy}{dx} = \frac{x+y+1}{2x+2y+3}$

17. A continuous function $f : R \rightarrow R$ satisfy the differential equation

$$f(x) = (1 + x^2) \left[1 + \int_0^x \frac{f^2(t)}{1+t^2} dt \right] \text{ then the value of}$$

$f(-2)$ is:

(1) 0

(2) $\frac{17}{15}$

(3) $\frac{-17}{15}$

(4) $\frac{15}{17}$

18. The solution of differential equation

$$\left(3x^2 \sin \frac{1}{x} + y\right) dx = x \cos \left(\frac{1}{x}\right) dx - x dy \text{ is}$$

(where, c is arbitrary constant)

(1) $\sin \left(\frac{1}{x}\right) = xy + c$

(2) $x^3 \sin \left(\frac{1}{x}\right) + xy = c$

(3) $x^3 \sin \left(\frac{1}{x}\right) = xy + c$

(4) $\sin(x) = x^3y + c$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (1)
2. (3)
3. (2)
4. (4)
5. (1)
6. (4)
7. (2)
8. (3)
9. (3)
10. (1)

11. 6
12. (3)
13. (2)
14. (3)
15. (4)
16. $x + y + \frac{4}{3}$
 $= ce^{3(x-2y)}$
17. (4)
18. (2)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>